

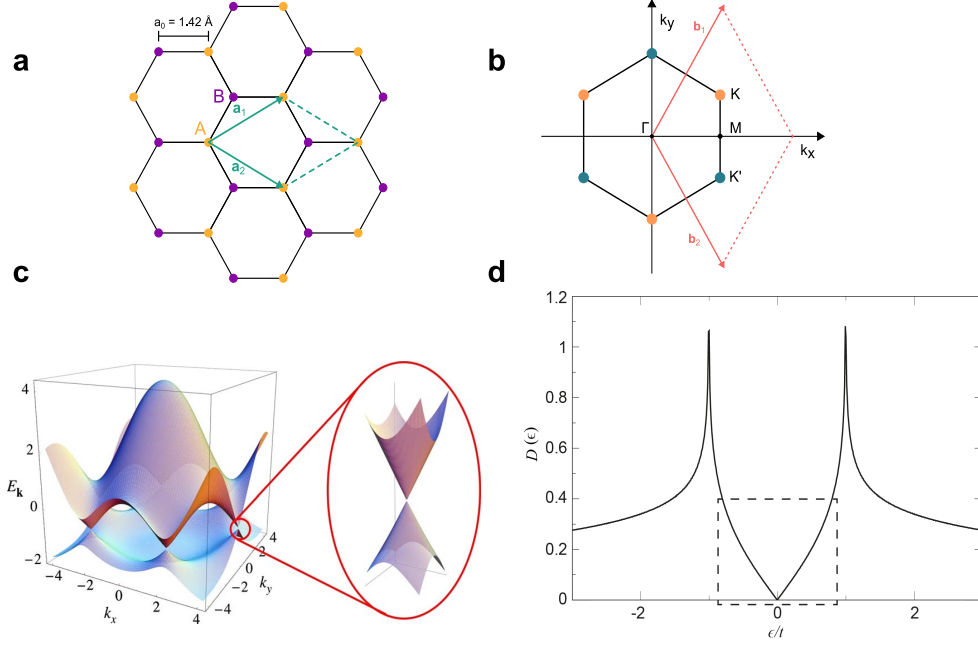


Figure 1.2: **a**, Honeycomb lattice of graphene. Two non-equivalent carbon atoms A and B in the primitive cells (direct space) are denoted in yellow and purple. The set of possible translations are showed by the lattice vectors $\mathbf{a}_1, \mathbf{a}_2$ (green arrows). **b**, The first Brillouin zone of graphene with vertices $\mathbf{K}$ (orange) and $\mathbf{K}'$ (green), and reciprocal-lattice vectors drawn in red. **c**, Graphene energy dispersion resulted from Eq. 1.11 (in units of nearest neighbor hopping parameter t). The magnification shows one of the Dirac cones, which are located at the $\mathbf{K}$ and $\mathbf{K}'$ points, emphasising the linear dispersion relation. **d**, Density of states as a function of t , showing pronounced van Hove singularities for $\epsilon \simeq \pm t$. The dashed rectangle indicates the low-energy linear dispersion region which is typically probed in back-gated transport experiments. Adapted from [6].

$$\mathcal{H}_{\text{TB}} = -t \sum_{\langle ij \rangle, \sigma} (a_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.}) - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} (a_{\sigma,i}^\dagger a_{\sigma,j} + b_{\sigma,i}^\dagger b_{\sigma,j} + \text{H.c.}), \quad (1.11)$$

The notation $\langle ij \rangle$ means the sum runs over pairs of electrons which are nearest neighbours; similarly, the notation $\langle\langle ij \rangle\rangle$ specify a summation over the next nearest neighbours.

The two operators $a_{i,\sigma}^\dagger$ ($a_{i,\sigma}$) and $b_{i,\sigma}^\dagger$ ($b_{i,\sigma}$) creates (annihilates) an electron with spin σ ($\sigma = \uparrow, \downarrow$) for atoms on sublattice A and B, respectively. The value of t which reproduces best experimental results is $t \simeq 2.8$ eV, and quantifies the nearest-neighbour hopping energy; similarly, t' denotes the next nearest-neighbour hopping energy.

By diagonalising this Hamiltonian it is possible to work out the energy dispersion of graphene, which reads

$$E_{\pm}(\mathbf{k}) = \epsilon_p - t'f(\mathbf{k}) \pm t\sqrt{3 + f(\mathbf{k})}, \quad (1.12)$$

with $f(\mathbf{k})$ defined as:

$$f(\mathbf{k}) = 2 \cos(\sqrt{3}k_y a) + 4 \cos\left(\frac{\sqrt{3}}{2}k_y a\right) \cos\left(\frac{3}{2}k_x a\right). \quad (1.13)$$

Eq. 1.12 identifies an upper (π^*) band associated with the plus sign and a lower (π) band described by the minus sign, and these are symmetric with respect to the single-particle zero energy. The experimental estimate reported for next nearest-neighbour hopping energy is $t' \simeq 0.1$ eV thus the asymmetry introduced is small.

If the system is not doped, there will be one electron per carbon atom, resulting in two electrons per cell: the lower band will be full, while the top band will be empty. Therefore, the system results to be semi-metallic. When examining the trend between the points $\mathbf{K}$ and $\mathbf{K}'$, the peculiarities of graphene's dispersion become clear. By taking the positive components for the $\mathbf{K}$ point as define in Eq. 1.7 and expanding the Eq. 1.12 as $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|\mathbf{q}| \ll |\mathbf{K}|$, one gets:

$$E_{\pm}(\mathbf{q}) = \epsilon_p \pm \frac{3at}{2}|\mathbf{q}| \quad (1.14)$$

This remarkable result tells us that the dispersion close to the Fermi level is linear in $\mathbf{q}$. For intrinsic graphene, at $T = 0$ K $E_F = \epsilon_p$. The Fermi velocity is defined by the usual relation $v_F = \frac{1}{\hbar} \left(\frac{dE}{d\mathbf{k}} \right)$, and in graphene $v_F \simeq 1 \times 10^6$ ms⁻¹ = 0.66 eV $\times$ nm $\times$ $\hbar^{-1}$.

An important consequence of this result is that whilst for a two-dimensional electron gas the density of states (DOS) is constant, in graphene close to the Fermi level the DOS reads:

$$\rho(E) = \frac{2A_{\text{cell}}}{\pi} \frac{|E|}{v_F^2} \quad (1.15)$$

If graphene is charge-neutral, there are two Van Hove singularities in the DOS at $\rho(\omega_{\text{VH}}) = \pm 1$ eV [8]. This behavior stands in stark contrast to the DOS of

carbon nanotubes and graphene nanoribbons as they exhibit $1/\sqrt{E}$ singularities due to their 1D structure, which imposes a quantization perpendicular to the longitudinal direction of the ribbon or tube.

Using the Einstein-Smoluchowski relation $D = k_B T / \eta = \mu k_B T$, which links an average friction coefficient η experienced when a flow of semiclassical charge carriers diffuse in a medium with coefficient D , the semiclassical conductivity may be computed from the DOS as:

$$\sigma_{\text{sc}} = e^2 \rho(E) v(E) \ell_{\text{el}} \quad (1.16)$$

where $v(E)$ is the carrier velocity, ℓ_{el} the elastic mean free path, and e the elementary charge. From Eq. 1.16 and the charge density $n(E)$, one can calculate the charge mobility, which, at zero temperature, reads:

$$\mu(E) = \frac{\sigma_{\text{sc}}}{en} \quad (1.17)$$

Close to the Dirac points, the experimental conductivities lie within $\simeq 2 - 5 \, e^2/h$ range, although several experiments have demonstrated a variation of more than an order of magnitude, owing to doping from the electrical contacts and/or the substrate [9, 10].

1.2.1 Graphene Transistors

The 2020 International Roadmap for Devices and Systems, which followed the 2017 International Roadmap for Semiconductors, places a significant emphasis on graphene in its study *beyond CMOS*, which investigates technologies and semiconductor materials that will supplement and/or replace Si (Tab. 1.1). Since carrier mobility is a well acknowledged characteristic for addressing carrier transport in nanoscale field-effect transistor (FET), the race to scale down transistor density per chip continues to motivate significant efforts to introduce high-mobility channel materials.

Initial interest for using graphene as a channel material in CMOS logic may have been fueled by its remarkable mobility [11]. In a standard CMOS gate, carriers are driven by a longitudinal electric field $\mathcal{E}_{\text{SD}}$ from the the source to the drain, and their

	Direct Gap			Indirect Gap		
	InP	Si	Ge	GaAs	InAs	Graphene
a_{lattice} (Å)	5.87	5.43	5.65	5.65	6.06	2.46
E_g (eV)	1.35	1.12	0.66	1.42	0.36	0
μ (cm ² V ⁻¹ s ⁻¹) at 300 K	2800	1400	3900	4600	16,000	200,000
v_{sat} (10 ⁷ cm s ⁻¹)	2.2	1	0.6	2.2	4.0	>5
DOS (m^*/m_0)	0.077	1.08	0.56	0.067	0.023	0
ϵ_r	12.4	11.9	16.0	13.1	14.6	2.4
κ (W m ⁻¹ K ⁻¹)	68	150	60.2	46	27	5000

Table 1.1: Comparison between graphene with the most common semiconductor electronics materials. The general interest in finding solutions for circumnavigating the absence of bandgap in graphene is due to various graphene characteristics, including mobility, carrier velocity at saturation, low electrostatic shielding provided by a very low dielectric constant, and an excellent ability to conduct heat, as shown by thermal conductivity values. The last point is particularly pertinent to the design of graphene-based thermal sinks.

motion is often limited by various scattering sources, such as collisions between electrons and vibrational modes of the crystal lattice from which the gate is made. In their first graphene-based FET prototype, Novoselov and colleagues measured an impressive intrinsic mobility μ_e^{in} at 290 K close to $\mu_e^{\text{in}} \simeq 10,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ [12] using a single layer of graphene on a Si/SiO₂ substrate serving as a back-gate. These results were explained by first assuming that graphene possesses conical rather than parabolic electron energy bands close to the Fermi energy, with electrons travelling at speeds close to the speed of light and, thus, with negligible effective mass. And second, by recognising that the honeycomb lattice symmetry implies a two-component wave function, which has the practical effect of cancelling direct carrier backscattering. Although it was believed that charged impurities [13] or microscopic ripples [14] scatterings were predominant at low temperatures, it is now

widely accepted that the very little electron–phonon scattering contribution even at room temperature is responsible for mobility values $\mu^{\text{in}} \simeq 200,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ [15].

In order to successfully realise a field-effect transistor, however, a suitable material has to be able to switch on and off, as these are mapped onto the two logic levels high-voltage/low-voltage, and been able to perpetuate this states against voltage variation. The figure of merit associated with this requirement is the so called $I_{\text{on}}/I_{\text{off}}$. On-off ratio quantifies the ability of any circuit component in electrical contact with the transistor to distinguish between its on and off states in the absence of gate voltages. A high on-off ratio corresponds to a low leakage current, resulting in an enhanced device performance.

While the scaling of a graphene-based transistor could be, in principle, be robust against off-state-current increases when the channel length is scaled down [16]. Klein tunnelling in the setting of graphene p–n junction results from the absence of backscattering for electrons belonging to the same Dirac cone and the conservation of momentum parallel to a straight p–n interface [17, 18]. Since incident electrons must conserve their transverse momentum, $k_y = 0$ and are barred from scattering backwards, they will tunnel through symmetric potential barriers with transmission probability of one.

Within the thermoionic emission model for the drain current, when graphene is placed on top of a semiconductor structure, the two work functions align accordingly based on the doping and carrier concentration levels (see a pictorial representation in Fig. 1.3). By assuming that the two work functions are identical, the height of the Schottky barrier between graphene and the semiconductor may be written as $\phi_B = E_g/2$, where E_g is the bandgap energy of the semiconductor, and the on-off ratio may be estimated from the simple relation [22]:

$$\frac{I_{\text{on}}}{I_{\text{off}}} \propto \exp\left\{\frac{E_g}{2k_B T}\right\}, \quad (1.18)$$

where the thermal energy $k_B T$ and the bandgap are both expressed in meV. Several researchers have seen bandgap values between 300 and 500 meV to yield the ideal ratios for CMOS applications [23, 24].

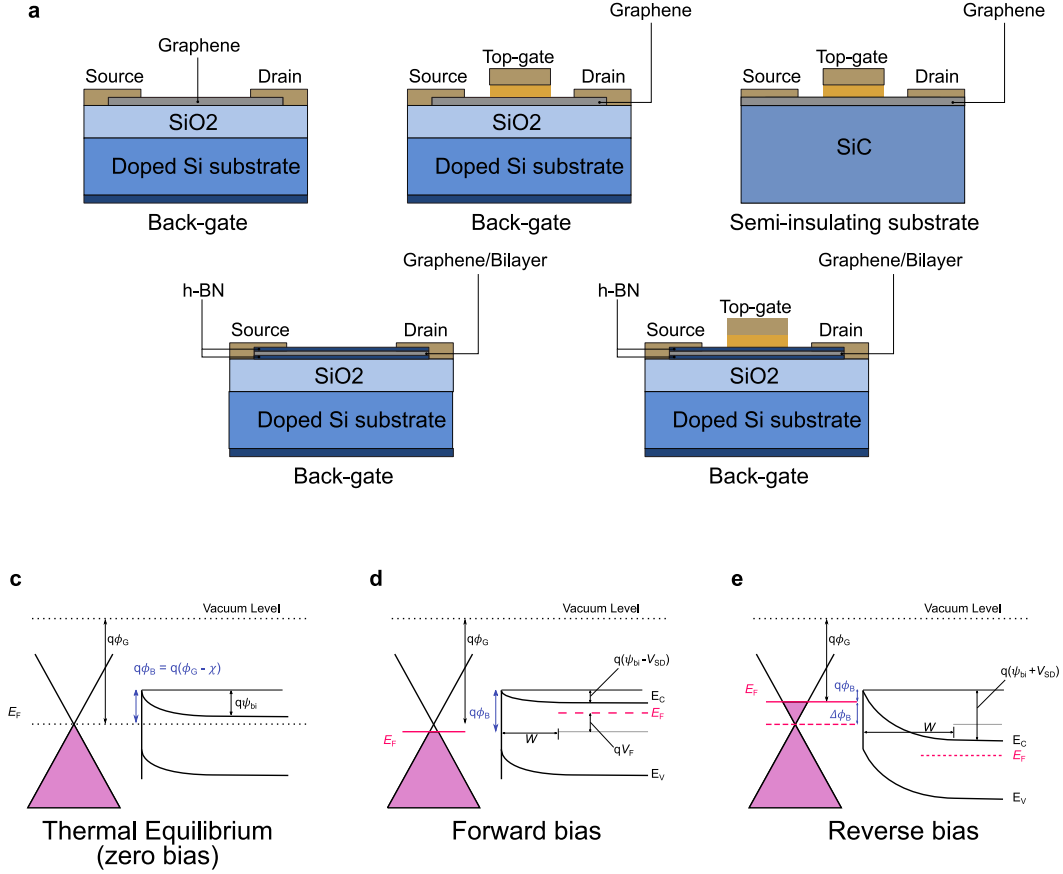


Figure 1.3: **a (top)**, Architectures of the main graphene MOSFETs. Charge carriers can be modulated in a graphene channel (epitaxial, CVD-grown, or multi-layer) connecting two metal contacts by either back-gate or top-gate. SiC substrate have been used for growing in-situ thermally decomposed graphene, which has found successful radiofrequency applications. **a (bottom)**, In recent years, several solutions have envisioned graphene substrate sandwiched between two hexagonal boron nitride (hBN) layers. Having a slight lattice mismatch (about 1.5% [19]) relative to graphene, $\epsilon_{\text{h-BN}} \simeq 3 - 4$ and significant bandgap (5.9 eV), it is considered the best insulating gate layer for graphene nowadays. Devices based on h-BN-sandwiched graphene have been testbeds for recent groundbreaking results such as unconventional superconductivity [20] and Stoner ferromagnetism [21]. **b**, Energy-band diagrams of graphene-semiconductor (G/S) contacts in thermal equilibrium, when a forward **c**, and reverse **d**, bias is applied. Symbols are associated to physical observables as explained in the main text.

Despite considerable efforts, the metallic nature of graphene has hindered CMOS architectures from reaching an off state with a high resistance, deeming them unsuitable for logic applications. In contrast, graphene-based electronics have found broad use in applications requiring high frequencies. For analogue applications, properties such as ballistic transport, linear current-voltage characteristics, high transconductance, and exceptionally high sustained currents ($> 10^8 \text{ A cm}^{-2}$) have proven to be optimal [25, 26].

The assessment of two transistor characteristic frequencies, namely cutoff f_T and maximum of oscillations f_{max} , is the primary tool to determine frequency response. The former indicates how quickly a signal may travel from the gate to the drain of the device and is often measured when the small-signal current gain approaches unity, whilst the latter is the highest possible frequency at which there is a non-vanishing power-gain [23]. The frequency response is inversely related to the length of the gate and proportional to the transconductance, which is approximately proportional to mobility[†]. Therefore, to achieve a high cut-off frequencies in a transistor, high carrier mobility must be maintained while the channel length is scaled down [16]. In this regard, numerous research groups have shown gapless graphene channel radiofrequency (RF) capabilities.

Late in 2008, it was claimed that the first device with a cutoff frequency in the gigahertz region was fabricated using exfoliated graphene placed on a high-resistivity silicon substrate, with a 30-nm-thick top-gate dielectric made of HfO_2 and a 500-nm-long gate [25]. Despite a reported cutoff frequency of about $f_T \simeq 14.7 \text{ GHz}$ it was stated that excessive contact resistances and gate length were the primary causes of low ($f_{\text{max}} < 1 \text{ GHz}$) maximum frequency performance. The following year, Pd/Au contacts and a 12-nm-thick aluminum oxide were implemented in a 350-nm-long graphene gate, and although the cutoff frequency was increased to 50 GHz the maximum frequency was increased only to 2 GHz [27]. This poor performance was attributed to the organic contaminations on the graphene surface resulting from

[†]This is strictly valid is the mobilities are described within the diffusive regime. Carriers are close to ballistic values, then the transconductance is constant.

the use of poly(methyl methacrylate) (PMMA) as a supporting layer during the transfer process, as well as the poor development of gate dielectrics [28].

After two years, a one-order-of-magnitude rise in the cutoff and the highest frequency of oscillation was demonstrated [29]. The devices are realised with an epitaxially grown silicon-carbide graphene MOSFET with a 240 nm gate length, Ti/Pd/Au contacts, and hafnium oxide 10 nm thick as top gate. Local gating is frequently preferred as gate-source, and gate-drain capacitances are inversely proportional to the channel length and minimised in this geometric configuration, hence improving the cut-off frequency that can be reached [30]. In addition to $f_T \simeq 100$ GHz, the highest maximum frequency ever measured was $f_{\max} \simeq 100$ GHz, beyond the capabilities of any technology at the time. It has been noticed that since the maximum oscillation frequency is significantly associated to device layout and parasitic resistances arising from the metallic contacts, partial passivation of the metal contacts and a multi-finger gate layout would be viable strategies to increase this parameter [31].

Later the same year, a 144-nm-long gate device with $f_T \simeq 300$ GHz was developed using mechanically exfoliated graphene and $\text{Co}_2\text{Si-Al}_2\text{O}_3$ nanowire as self-aligned multi top-gates [32]. This result was also confirmed by comparing it to the predicted value based on the relationship[†] between dc transconductance, gate capacitance, and the cutoff frequency in the case where total input parasitic capacitance is negligible. The authors did not report the power $G_{\max}^{1/2}$, only the current gain $|h_{21}|$, making it difficult to assess the $f_{\max}$ of the device. A rough estimate might be extracted by looking at the results reported by Cheng and colleagues on a similar device where the gate was realised through a stack of Au/Ti/ Al_2O_3 [34]. The achieved enhancements in the cutoff frequency response ($f_T \simeq 427$ GHz for a channel length of 67 nm) compared with CVD graphene-based device with similar gate length were, however, eclipsed by minor improvements to the device's maximum frequency response up to 8 GHz when the gate length was lowered to 46 nm. As for the cutoff frequency, it is vital to minimise all parasitic resistances and capacitance in order

[†] $f_T = g_m/(2\pi C_g)$ [33]

to optimise $f_{\max}$. In addition, the output conductance must be decreased in order to obtain the highest feasible operating speeds, which is often accomplished by operating the transistor in the drain current saturation regime.

IBM did also some research on the saturation behavior of short channel graphene transistors leveraging CVD graphene transferred to diamond-like and SiC substrates in an attempt to improve the power gain [35]. These are particularly attractive possibilities since the substrates are commercially accessible on high-quality wafers, allowing the scaling up of graphene arrays quite straightforward. Since the work function of graphene ($\phi^g = 4.5$ eV) is greater than that of silicon carbide ($\phi^{4\text{H-SiC}} = 3.7$ eV), negative charges accumulate on the graphene side of the contact, leaving graphene n-doped [36]. The authors compensated this unwanted doping by using an innovative technique involving plasma-enhanced CVD silicon nitride as an intrinsic p-type gate dielectric. Positive performances at high speed were assessed not only by good cutoff frequencies ($f_T \simeq 350$ GHz), but also by a twofold increase in maximum oscillation frequency $f_{\max} \simeq 40$ GHz. According to my knowledge, the current graphene transistor power gain record of approximately 70 GHz was achieved by using a (000-1) carbon-terminated surface of silicon carbide instead of the conventional (0001) Si, indicating that a low contact resistance between the gold contacts and graphene is the primary factor responsible for the high maximum frequency [37].

In terms of cutoff frequencies, the performance of graphene-based devices is comparable to that of the leading solid-state competitors, such as GaAs pHEMT, InP HEMT, and Si MOSFET. The performance in terms of maximum frequencies, however, is one or two orders of magnitude below the required values for a competing technology [38–40].

In the scientific literature, one of the most commonly cited underlying factors for this behaviour is the weak drain saturation current in graphene, which ultimately depends on the lack of a bandgap in the channel. Because of their intrinsic semiconductivity, graphene-derived materials such as nanoribbons and carbon nanotube MOSFET could be a viable solution to this problem. In the following

sections, I will investigate some important aspects of the electronic response at the device level for each class of materials.

1.3 Carbon Nanotube

The carbon nanotube literature is vast and well-documented. In the part that follows, my primary aims are two. First, I will provide a comprehensive review of carbon nanotube field-effect transistors (CNTFETs), highlighting crucial elements that have, in my opinion, played a significant role in enhancing the competitiveness of nanotube-based transistors. The second objective is to quantify the device's performance in order to compare it to GNR-based transistors.

In contrast to graphene, CNT have an electrical bandgap that varies according to the nanotube's chirality and diameter [41]. Any nanotube can be defined in the real space by defining two unit vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ and any linear combination of these with integer coefficients, typically labelled m and n . Depending on their values, a nanotube may be metallic, quasi-metallic, or semiconducting; to this simplified picture, one may add exceptions induced by the strong curvature (which leads to a certain kind of spin-orbit interactions) when the diameter of the tube is extremely small [42].

As depicted in Fig. 1.4, states resulting from quantization in the circumferential direction may or may not pass through one of the $\mathbf{K}$ points in the first Brillouin zone of graphene. The resulting nanotube will be metallic if it crosses a $\mathbf{K}$ point; else, it will be semiconducting. This feature of the crystal structure was essential for the development of the first prototype CNTFET.

1998 saw the discovery of charge-carrier modulations within a nanotube produced by a gate-induced field from the group of Cees Dekkar [43]. These early FETs were manufactured using a doped silicon substrate as the gate to regulate the energy bands, with thermally-grown silicon dioxide acting as the dielectric between the nanotube and the substrate, and Au contacts to electrically contacting the nanotube.

Since then, other transistor designs have been proven and can be classified fundamentally as devices with or without a spacer. The length of the spacer L_{spr}

is the gap between the source/drain metal contacts and the channel. Given that a certain class of nanotube is intrinsically semiconducting, in spacer-less devices the motion of charge carriers is not induced by the electric field $\mathcal{E}_{SD}$ in the channel, but are instead injected from the contacts. The role of the gate is to lower the thermal barrier in the channel, thereby permitting tunnelling through the Schottky barrier (SB) at the source. In this case, the height of the metal's work function ϕ^M plays a crucial part in the transistor's performance.

In contrast, CNTFETs with spacers are less constrained by injection at the source since the electric field created by the gate between source and drain controls the carriers. It has been demonstrated that a device with spacers is more resistant to ambipolar conduction due to the substantial carrier barrier injection at the drain [44]; Tab. 1.2 provides an overview of gated-spacer (top and bottom) devices, whereas Tab. 1.3 reports the performance of a few devices based on the kind of electrical contact between the nanotube and the electrode.

Another critical issue for nanotube transistors is to control the conduction type of devices: *p*-type transfer characteristics have been reported for the vast majority of tubes. This is partly due to *p*-type of contaminations following the lithographic process that are involved in the conduction [45] even though *n*-type devices are indispensable for nanotube integrated circuits with low-power dissipation, similar to Si-based CMOS devices.

An interesting solution tested, for example, was doping the nanotube-channel surface with element of the first two groups of the periodic table (ions K^+ , Mg_2^+ , etc) to fabricate *n*-type nanotube FETs [46, 47]. Precise control of the doping density per nanotube unit length is necessary, which is problematic. Additionally, ionized impurities are often a source of uncontrolled electron transport due to scattering events with the charge carriers. Moreover, instability in the air, and temperature fluctuations are other factors which have discouraged the chemical doping route.

The operation mechanism of the nanotube FETs has been widely accepted to be satisfactorily explained by Schottky barrier modulation model, in which the gating action is dominated by the Schottky barrier formed at the contact between the

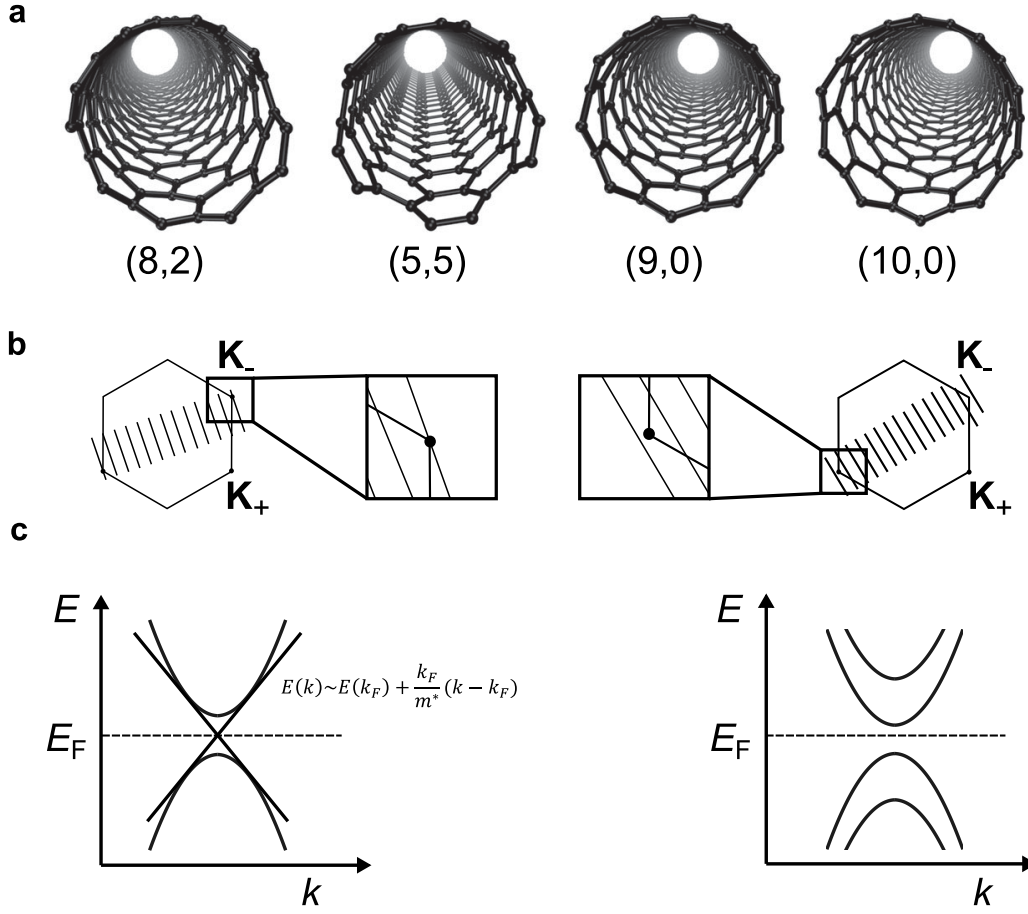


Figure 1.4: **a**, Cartoon structure of a chiral (8, 2), an armchair (5, 5), and zigzag (10, 0) nanotube. **b**, Graphene Brillouin zone and allowed k-vectors for the (8, 2) and (10, 0) CNTs. A metallic tube include one of the **K** points among the set of allowed k-vectors, as shown in **c**. It is then possible to linearise the energy in the momentum space for low-excitations (close to the Fermi energy). Conversely, the **K** point is not an allowed vector for semiconducting nanotube, and this leads to an energy gap. Inspired by [44].

nanotube and the source metal [48]. Solid-state semiconductors have dangling bonds at the surface, resulting in Fermi level pinning, which means that the variation of the band bending is independent of the work function of the metal, even when it is significant [49]. Since the ideal tube structure has no open bonds, there is no pinning of the metal Fermi level to mid-gap states, which justifies the study of various work functions for improving performance and tuning polarity. Normalized by the tube width. Numerous metals have been examined for nanotube interaction. It has been established that palladium ($\phi^{\text{Pd}} \simeq 5.2\text{--}5.6\text{ eV}$) creates a nearly perfect

p -channel contact with the valence band of CNT. The reason for this phenomena was explained within the Schottky barrier framework in which the larger work function makes the Schottky barrier high for holes lower, thus reducing the overall contact resistance between the metal and the nanotube [50]. This behaviour, however, is not universal as demonstrated by other metals such as platinum ($\phi^{\text{Pt}} \simeq 5.65 \text{ eV}$) which produces weaker contacts despite having a larger work function than Pd due to the physical wetting of the metal on the nanotube [51].

Taking these reports into account, if we use materials with a small work function as contact electrodes of nanotube FETs, the Schottky barrier height for electrons is low. In this case, it is expected that electrons are injected from the contact electrode into the nanotube and n -type nanotube FETs will be realized without any doping into the nanotube channel. The other problem associated with using low work function metals as contacts is their increased propensity to oxidise. Using metals with a high work function to cap the contact metals is a simple and effective solution for the contact surface's tendency to destabilise in air. High-performance n -type devices were demonstrated by using Sc- and Y-contacted CNTFETs, delivering high on state currents (250 micro ampere), mobility of $5100 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, and drain induced barrier lowering (DIBL) of 105 mV V^{-1} [52]. These devices reported an impressive performance for the subthreshold swing $SS \simeq 73 \text{ mV dec}^{-1}$.

The subthreshold swing measures the rate at which the gate voltage switches off the transistor, and it is the gate-voltage shift necessary to cause a one-order-of-magnitude change in drain current. It is a transistor' figure of merit often indirectly linked to heat dissipation, as this represents of the primary obstacles to miniaturising CMOS technology. Since the carriers are thermally injected, it is challenging to reduce the operating voltage while improving switching (operational) speed and maintaining a low off-state power dissipation. Since the off-state power is proportional to the off-state current, $P_{\text{off}} = RI_{\text{off}}^2$, the objective is to achieve a low off-state current without sacrificing the on-state output current. The process of charge carrier transfer by thermal means imposes a limit on the subthreshold

Table 1.2: Figures of merit for a few CNFET reported in the literature. SS : subthreshold swing; g_m : intrinsic transconductances; L_{ch} : channel length; μ : carrier’s mobility; f_T : cutoff frequency; f_{max} : maximum of oscillations frequency

<i>Device type</i>	I_{on}/I_{off}	SS (mV/dec)	g_{mt} ($\mu S cm^{-1}$)	L_{ch} (nm)	f_T [f_{max}] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[56]
	10^5	~ 85	40	15	—	[57]
	10^5	~ 85	20	20	—	[57]
	10^5	60	20	200	—	[58]
	10^5	~ 63	—	200	—	[59]
	10^4	80	50	240	—	[60]
	—	—	143	800	[1]	[61]
	10^4	—	—	1,500	1.5 – 3.0	[62]
	—	70	$12^\dagger$	2,000*	—	[63]
	10^8	—	—	$< 10,000^*$	—	[64]
Top gate	—	—	- 500	—	30	[65]
	10^6	—	$20^\dagger$	80	—	[66]
	2	—	~ 50	100	25	[67]
	50	—	310	100	~ 70 [80]	[68]
	—	—	0.3	210	—	[69]
	10^5	—	33.5	85,000	—	[70]

† Normalized by the tube width.

1 Referred to the gated length of a nanotube.

2 For the n -type transistor, whereas 10 μS were measured for the p -type.

slope, which for the metal–oxide–semiconductor field-effect transistor (MOSFET) technology is approximately $2.3k_B T/q = 60 \text{ mV dec}^{-1}$ at room temperature.

One of the most promising approaches to overcoming the Boltzmann limit proposes a different source–carrier injection mechanism based on Zener tunneling rather than thermal means. When the transmission probability does not depend on the energy gap between the metal and the nanotube work functions, but instead occurs through interband tunneling, the transmission probability becomes temperature independent. Several architecture proposals have envisioned carbon nanotubes and graphene nanoribbons transistors operating on this basis [53–55].

Table 1.3: Comparative simulation for the four major type of CNFETs [71]. These results are obtained by setting $V_{SD} = 0.5$ V.

<i>Device type</i>	I_{on} (A)	I_{off} (A)	I_{on}/I_{off}	g_{mt} (μ S)	SS (mV/dec)
Conventional	1.41×10^{-5}	4.06×10^{-10}	3.4×10^4	29	99.6
Partial S/D-doped	1.50×10^{-5}	2.33×10^{-8}	6.4×10^2	63.6	93.1
Schottky-Barrier	3.26×10^{-6}	2.31×10^{-8}	1.41×10^2	19.6	109.8
Tunnel	5.53×10^{-6}	1.16×10^{-8}	4.76×10^2	13	107.9

1.3.1 Scaling

Si technology's ability to function below 10-nm channel length was questioned for a long time, but recent scientific advances have demonstrated its viability and fostered a plethora of commercial uses (currently, modern laptops employ the 5-nm technology). Nonetheless, Si scalability has always been a significant starting point for investigating methods and materials that might overcome short-channel effects (threshold voltage roll-off, off-state leakage current, and channel length modulation are just a few of the problematics).

In carbon nanotube transistors, the difference between two key lengths for scaling, namely the contact L_c and channel L_{ch} lengths, is crucial yet often underestimated. Aggressive scaling for the channel length typically results in the loss of control over the band bending across the gate: intuitively, as the channel length is reduced, the electric field created by the source contact will be felt by the charge carriers more and more. The thesis presented here is that carbon nanotubes provide a suitable platform for such scaling due to their ultrathin hollow nature. Alternatively, this could be a double-edged sword. Due to a higher tunnelling probability, the lower effective carrier mass of nanotubes[†] may increase the tunnelling current from source to drain; typically, this effect has implications such as an increase in the subthreshold swing and leakage currents in the off state.

[†] $m_{CNT}^* = (0.228 \pm 0.129)m_0$ [74]

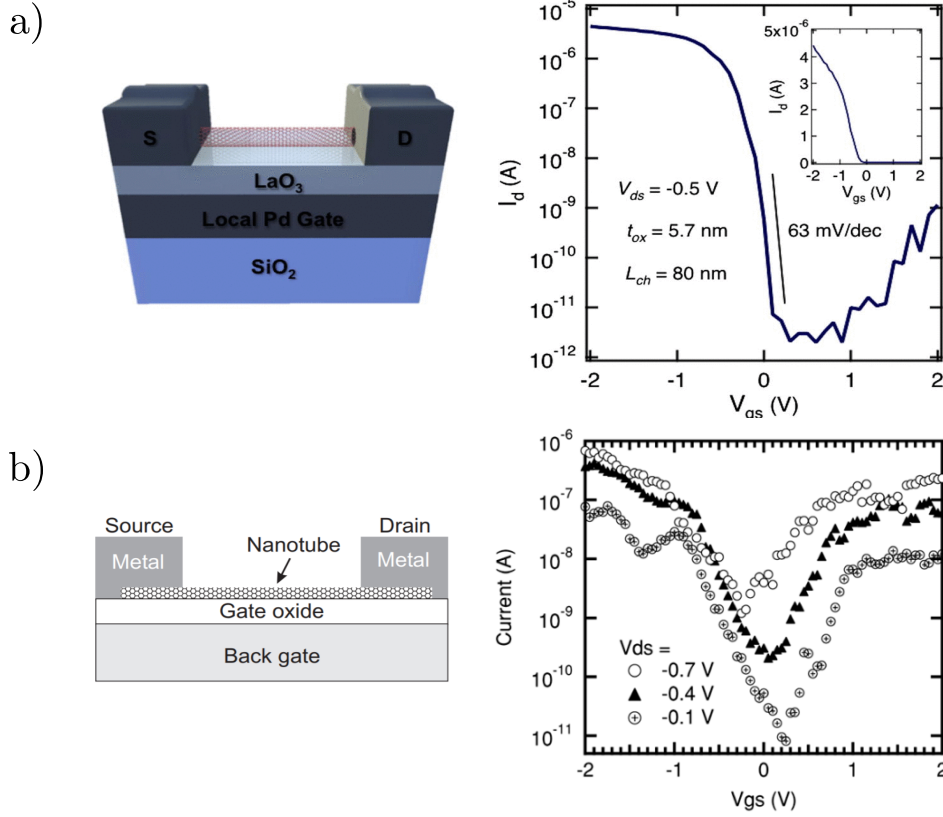


Figure 1.5: (a) Lanthanum oxide is a high- κ dielectric thermally stable when placed in contact with Si. With these devices it has been possible to obtain subthreshold swing close to the theoretical limit of $SS = 60 \text{ mV dec}^{-1}$ [72]. (b) Silicon dioxide back-gated architecture. In the more traditional version of these devices, ambipolar features are often present due to the conduction through both electrons and holes. The off current exhibits a strong dependence from the source-drain bias as—when it is lowered— I_{off} increases exponentially and the corresponding gate voltage shifts to more negative values.[73]

Franklin and coworkers conducted one of the earliest systematic studies on channel length scaling effects in 2010 via a series of experiments on nanotubes using hafnium-based oxide-coated local bottom gates [75]. When the channel length was reduced by more than two orders of magnitude (from $3 \mu\text{m}$ down to 15 nm), a consistent trend was found in both the threshold voltage ($V_{\text{Th}} \simeq (-0.5 \pm 0.2) \text{ V}$) and subthreshold swing ($SS \simeq 90 \text{ mV dec}^{-1}$) values.

The threshold voltage is the gate voltage required to fully invert the semiconductor. It is typically measured by applying a slight drain bias and measuring the

drain current as a function of the gate voltage, provided that the drain current is at least one order of magnitude smaller than the gate voltages applied, scaled accordingly by the gate capacitance.

In addition, it was predicted that scaling had little to none effect on the transconductance g_m of the device. The transconductance of a conventional MOSFET is the change in drain current divided by the amount that the gate-to-source voltage varies at a constant bias. Typical values lies in the range (1–30) mS μm^{-1} for a small-signal field-effect transistor.

For an n -channel MOSFET with a certain threshold voltage, the drain current at a given gate in the saturation regime ($V_{\text{Th}} > V_G - V_{\text{SD}}$) is defined as follows:

$$I_{\text{SD}} = \frac{1}{2} \mu_n C_{\text{ox}} \frac{W}{L} (V_G - V_{\text{Th}})^2, \quad (1.19)$$

where μ_n is the electron mobility, C_{ox} is the gate oxide capacitance, W and L are the gate width and length, respectively.

It is possible to work out the value of g_m by using the Shichman–Hodges model and by assuming that the threshold voltage V_{Th} is a device constant, such that $V_{\text{ov}} = V_G - V_{\text{Th}}$:

$$g_m = \frac{\partial I_{\text{SD}}}{\partial V_{\text{GS}}} = \frac{2 I_{\text{SD}}}{V_{\text{ov}}}, \quad (1.20)$$

This quantity is the so-called overdrive voltage, which corresponds physically to a value just over the threshold voltage. this quantity is often used in the designing phase of a transistor and as it elegantly links the drain current with a voltage: in fact, by rearranging and identify one gets:

This quantity V_{ov} is the so-called overdrive voltage, which physically corresponds to a value just above the threshold voltage, and it is typically defined as $V_{\text{ov}} = V_G - V_{\text{Th}}$. This parameter is frequently employed during the design phase of a transistor because it conveniently ties the drain current with the applied bias: in fact, by rearranging Eq. 1.19 and identifying $k_n = \mu_n C_{\text{ox}} \frac{W}{L}$, one obtains:

$$I_{\text{SD}} = \frac{1}{2} k_n V_{\text{ov}}^2 \implies V_{\text{ov}} = \sqrt{\frac{2 I_{\text{SD}}}{k_n}} \quad (1.21)$$

The advantage of this equation is that the relationship written in terms of overdrive voltage rather than threshold voltage holds true for both n - and p -types of doping. To provide some background, the overdrive voltage for 65 nm technology is commonly set between 70 and 200 mV. Franklin and his colleagues produced transconductance values as high as $40 \mu\text{S} \mu\text{m}^{-1}$ for a CNTFET with a channel length of 15 nm, which corresponds to transconductance values between approximately (91–110) $\text{mS} \mu\text{m}^{-1}$, thereby meeting the requirements for compatibility with Si technology.

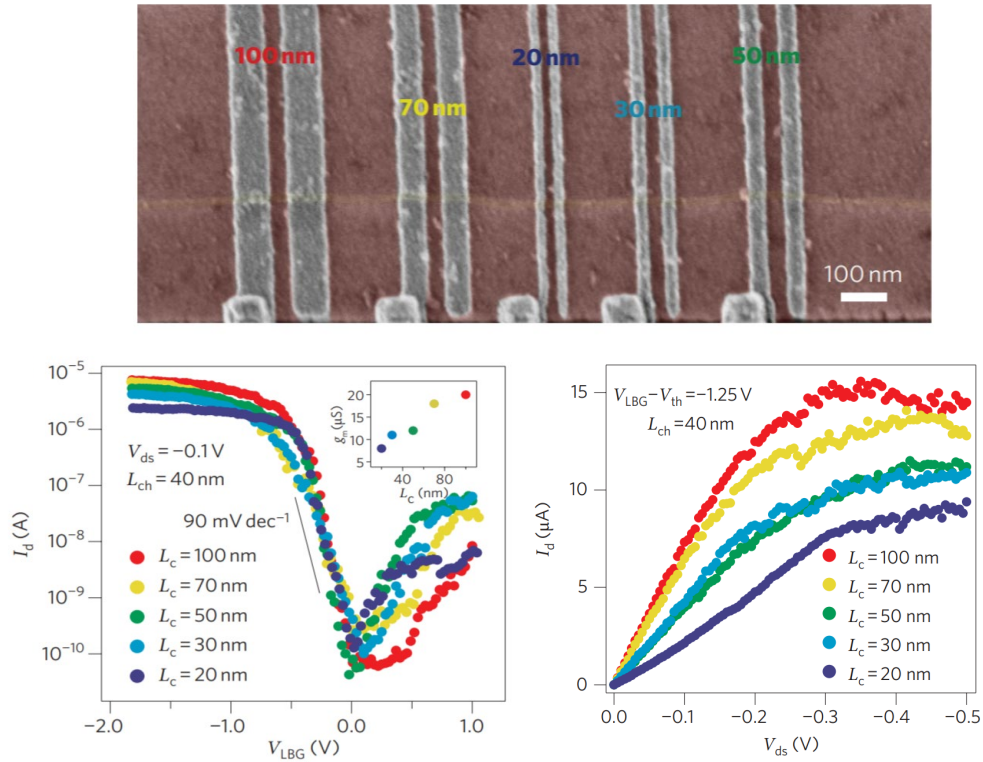


Figure 1.6: Top frame is the false-coloured SEM image of several transistors on the same nanotube but having different L_c , from 100–20 nm. Even for the smaller contacts the gate has complete control over the channel and the device switches quicker than most Si MOSFET-based competitors. Small contact lengths, however, severely degrades the on-state current, although the saturation regime it is still well within the experimental accessible region. Adapted from [75]

Two years after their initial breakthrough, Franklin and his colleagues shrunk the channel length of a nanotube transistor to less than 10 nanometers [56]. Their

technique involved the use of smaller tubes and a reduction of the necessary effective oxide thickness (EOT) to 0.65 nm. When the gate oxide is excessively thick, even by using local gates, the gate tends to lose control over the flow of carriers in the channel. The length of the channel was successfully lowered from 320 to 9 nm. The authors reported on-current with ballistic values for transistors with channel lengths close to 41 nm. By examining the subthreshold swing of $SS \simeq 94 \text{ mV dec}^{-1}$, one may appreciate the greater control from the gate to the conductive channel, which at the time exceeded any silicon-based device. The author advocated decreasing the nanotube's diameter to reduce the subthreshold swing and off-current with minimum influence on the on-current. The process of scaling down a CNTFET does not come without any costs, as the contribution of carrier back injections to the drain's conduction band grows as gate length decreases. Typically, this is indicated by a minimum current rise. To avoid this unfavorable impact on the off-state, the nature of the junction contact and its scaling behaviour must be examined.

The authors reported an increase on the minimum current as L_{ch} decreases from 320 nm to 9 nm, attributed to a back-injection of the carriers into the conduction band from the drain, while the on-current approached the ballistic value for $L_{\text{ch}} \leq 41 \text{ nm}$ [75]. Impressive switching behaviour, with sub-threshold swing (SS) about $SS \simeq 94 \text{ mV dec}^{-1}$, was measured, thus far outperforming any silicon-based of that time. based on previous results [75], the authors claimed the possibility of lowering the SS and the off-current at minimal cost to I_{on} by reducing the diameter of the nanotubes.

1.3.2 Contact Length Scaling

MOSFET devices, when shrunk down to the nanoscale, need to consider not only L_{ch} but also the length of the nanotube with the source and drain, L_{c} . In particular, computational studies have shown that higher contact transparency is expected when the metal-nanotube coupling is weak [76–78]. Especially for very long contacts, one conclusion which could be drawn from these studies is that in order to achieve high contact transparency one should minimize coupling